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A new species of *Laccaria* in montane cloud forest from eastern Mexico

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ABSTRACT

An undescribed species of *Laccaria* was discovered in the Santuario del Bosque de Niebla of Xalapa, Mexico, in a montane cloud forest preserved under the protection of the Instituto de Ecología A.C. in Veracruz State. This new species is distinct based on basidiome morphology and supported by phylogenetic analyses of sequences of the internal transcribed spacer (ITS) and nuclear large subunit (nLSU) of the ribosomal RNA gene. Thirteen different collections obtained during 2012–2014, provided documentation of the broad morphological variation and confirmed the diagnostic color changes of this species. It is phylogenetically associated with Metasection *Amethystina* but lacks violet pigments in the mycelium and stipe base that are characteristic for species placed in that Metasection. Its relationship to other taxa in *Laccaria* is not obvious at this time. Descriptions, color images of the basidiomata, scanning electron photomicrographs of basidiospores and comparisons with similar taxa are presented.

1. Introduction

Laccaria is a cosmopolitan ectomycorrhizal genus, associated mainly with Fagaceae and Pinaeae (Mueller 1992). Some members of this genus have been considered important fungal symbionts useful in forest agriculture and regeneration, and thus have been the subject of numerous research efforts for in vitro establishment of ectomycorrhizal symbiosis (e.g. Shaw and Molina 1980; Molina 1980, 1982; Molina and Palmer 1982; Hung and Trappe 1983; Molina and Trappe 1984; Hung and Molina 1986a, b; Richter and Bruhn 1989; Pera et al. 1998). More recently the genome of *L. bicolor* was entirely sequenced (Martin et al., 2008) allowing recognition of gene sets involved in rhizosphere colonization and symbiosis, which leads to a better understanding of this process in nature.

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In Mexico the first taxonomic study of *Laccaria* was produced by Aguirre-Acosta and Pérez-Silva (1978) and nine taxa were reported. Succeeding contributions to the knowledge of *Laccaria*, included additional new records (Montoya et al. 1987; Cifuentes et al. 1990) or studies of ectomycorrhizae formed by some *Laccaria* species in culture conditions (Santiago-Martínez et al. 2003; Carrasco-Hernández et al. 2010) and even information about traditional use of some species as food in some Mexican communities (e.g. Montoya et al. 2002, 2003; Lampman 2007; Pérez-Moreno et al. 2008).

During our monitoring program in the Santuario del Bosque de Niebla, a montane cloud forest ecosystem under the protection of the Insituto de Ecología A.C. (INECOL), we discovered in 2012 what appeared to be a white species of *Laccaria*. Because of the highly unusual white basidiomata, the mycelial mats of this taxon were subsequently monitored during Jun to Oct 2013 and May 2014, in order to obtain a series of fresh collections to document the extent of macro- and micromorphological variation of this species. Additional collections confirmed the extreme color changes of the basidiomata. The unique molecular signature of this species, based on two nuclear ribosomal RNA genes, also supports recognizing it as a new.

2. Materials and methods

2.1. Morphology: sampling and documentation

Collections of *Laccaria* were made on eight visits conducted between Jul 2012 and May 2014. Preserved collections are kept in the herbaria at the State University of New York College at Cortland (CORT) and the Instituto de Ecología, A.C. (XAL), (acronyms for herbaria from B. Thiers, continuously updated; Index Herbariorum: http://sweetgum.nybg.org/ih/).

Macromorphological features and colors were recorded from fresh basidiomes. Colors were described using Kornerup and Wanscher (1967, 1978) and Munsell Soil Colour Charts (1994). Micromorphological study tissues utilized dried specimens, after rehydration in alcohol and water, then mounting in 3% aqueous potassium hydroxide solution or Melzer's reagent. Cotton blue was used to determine cyanophily. The protocols of Montoya and Bandala (2003) were followed for calculating spore size ranges. Thirty-five basidiospores per collection were measured (length and width of the sporoid excluding the ornamentation) and given as a range with the symbol X representing mean values. Q represents the basidiospore length/width ratio and is given as range of mean values. Line drawings were made using a drawing tube attached to microscope. The SEM images were obtained after critical point drying of pieces of lamellae previously rehydrated in ammonia, fixed in glutaraldehyde and dehydrated in an ethanol series (Bandala and Montoya 2000).

2.2. Molecular phylogeny

2.2.1. DNA extraction, PCR amplification, and sequencing

Genomic DNA was extracted according to Montoya et al. (2014). PCR was performed to amplify the internal transcribed spacer (ITS) 1, 5.8S, ITS 2 and large subunit of 28S ribosomal RNA gene (nLSU), using primers *ITS1F, ITS5/ITS4, LrOR/LR21, LR7* (White et al. 1990; Gardes and Bruns 1993). PCR conditions, as well as procedures for the purification of amplified PCR products, cycle sequencing reactions and their purification were done according to Montoya et al. (2014). Once sequences were assembled and edited, they were deposited at GenBank (http://www.ncbi.nlm.nih.gov) (Table 1).

2.2.2. Phylogenetic methods

Sequences obtained in this study were combined with sequences of *Laccaria* species retrieved from GenBank (http://www.ncbi.nlm.nih.gov/) (Table S1) that had previously been analyzed (Osmundson et al. 2005; Wilson et al. 2013), and two datasets (ITS and LSU) were constructed in PhyDE v.0.995 (Müller et al. 2006) and aligned using Muscle (Edgar 2004). A third combined (ITS + LSU) set was assembled in PhyDE v.0.995 and uploaded in TreeBASE (http://purl.org/phylo/treebase/phylows/study/TB2:S16000?x-access-code=f852789b6028e33262d452aca811df7b&format=html). All operational taxonomic units in this dataset contained both ITS and LSU sequences. A phylogeny was tested under maximum likelihood (ML) and Bayesian inference (BI). The best evolutionary model for the combined dataset was calculated with Mega 6.06 (Tamura et al. 2013). For ML analysis Mega 6.06 with 500 replicates for bootstrap was used. BI analysis using MrBayes v 3.2.1 (Ronquist et al. 2012) was produced using two independent runs of four chains for 10,000,000 generations, sampling one tree every 1,000 generations. Sample points collected as prior stationarity (convergence of likelihood scores) were eliminated as the burn-in (10%). Posterior probabilities for supported clades were determined by a 50% majority-rule consensus of the trees retained after burn-in. *Psathyrella rhodospora* and *Cortinarius violaceus* were used as outgroups, following Wilson et al. (2013). The phylogenies from ML and BI analyses were displayed using Mega 6.06 and FigTree v 1.3.1 (Rambaut 2009) respectively.

3. Results

3.1. Phylogenetic analyses and phylogeny

The combined (ITS + LSU) dataset included 76 sequences (38 from ITS and 38 from LSU), of which 10 were newly generated (5 from ITS and 5 from LSU), with accession numbers indicated in Table 1. The phylogeny was inferred by using the ML method based on the Tamura 3-parameter model (Tamura 1992). Initial tree(s) for the heuristic search were obtained automatically by applying Neighbor-Join and BioNJ algorithms to a matrix of pairwise distances estimated using the Maximum Composite Likelihood (MCL) approach, and then selecting the topology with superior log likelihood value. A discrete Gamma distribution was used to model evolutionary rate differences among sites (5 categories (+G, parameter = 0.1427)). All positions containing gaps and missing data were eliminated. There were a total of 1,022 positions in the final dataset. The tree with the highest log likelihood (−2,978.1057) is displayed in Fig. 1, which includes the bootstrap values, as well as the posterior probabilities. The

Table 1 – *Laccaria* taxa used in this study: samples, location and GenBank accesion number for ITS and LSU sequences.

Taxon	Voucher specimen	Location	GenBank	
			ITS	LSU
L. alba	AWW438	China, Yunnan, Shangrila Co.	JX504094	JX504178
L. alba	GMM6131	China, Bai Shan, plots near electric Dam	JX504131	JX504210
L. alba	F1121461	China	JX504129	JX504209
L. amethystina	GMM7621	France, Forestcomaniale de Ste. Croix	JX504150	JX504224
L. amethystina	GMM7633	France, Forestcomaniale de Ste. Croix	JX504154	JX504228
L. bicolor	AWW537	USA, Illinois, Johnson Co.	JX504105	JX504189
L. bicolor	AWW594	USA, Alaska	JX504114	JX504197
L. laccata	GMM7606	France, Vallon d. Arcillac	JX504147	JX504221
L. aff. *montana*	AWW444	China, Xizang (Tibet), Bomi Co.	JX504095	JX504179
L. aff. *montana*	AWW446	China, Xizang (Tibet), Bomi Co.	JX504097	JX504181
L. aff. *montana*	GMM7630tibet	China, Xizang (Tibet)	JX504151	JX504225
L. nobilis	F1120629	China	JX504124	JX504204
L. ochropurpurea	PRL3777	USA, Illinois	JX504169	JX504246
L. ohiensis	AWW545.	USA, Illinois, Johnson Co.	JX504106	JX504190
L. proxima	GMM7631	France, Forestcomaniale de Ste. Croix	JX504152	JX504226
L. proxima	GMM7596	France, Vallon d. Arcillac	JX504142	JX504217
L. pumila	GMM7637	France, Fraychirede Bog	JX504156	JX504229
L. roseoalbescens[a]	LM5042	Mexico, Xalapa, Veracruz	KJ874327	KJ874330
L. roseoalbescens[a]	LM5099	Mexico, Xalapa, Veracruz	KJ874328	KJ874331
L. roseoalbescens[a]	PG3	Mexico, Xalapa, Veracruz	KJ874329	KJ874332
L. roseoalbescens[a]	VB4677	Mexico, Xalapa, Veracruz	KJ590508	KJ590511
L. roseoalbescen[a]	VB4678	Mexico, Xalapa, Veracruz	KJ590509	KJ590510
L. salmonicolor	GMM7602	China, Xizang (Tibet)	JX504145	JX504220
L. salmonicolor	GMM7596tibet	China, Xizang (Tibet), Linzha Co., Highway	JX504143	JX504218
Laccaria sp.	HKAS42577	China, Yun Co., Houqing Town, Qinshan	JX504158	JX504234
Laccaria sp.	HKAS53725	China, Sichuan, Jinchuan	JX504164	JX504239
Laccaria sp.	SB2067	Portugal	JX504171	JX504248
Laccaria sp.	AWW251	Malaysia, Pahang Prov., Frazier's Hill	JX504093	JX504177
Laccaria sp.	GMM6784	China, Danetangyakou	JX504135	JX504211
Laccaria sp.	F1120731	China	JX504125	JX504205
Laccaria sp.	GMM7601	France, Vallon d. Arcillac	JX504144	JX504219
Laccaria sp.	F1120374	China	JX504123	JX504203
Laccaria sp.	F1121424	China	JX504127	JX504207
L. trichodermophora	GMM7733	USA, Texas	JX504157	JX504230
L. trullisata	PRL7587		JX504170	JX504247

[a] Samples and sequences obtained here.

phylogeny supports recognition of this morphologically distinct Mexican *Laccaria* with sequences clustering in a strongly supported (BS = 100, BPP = 1) and isolated clade (Fig. 1). A second phylogeny (see Supplementary Fig. S1), produced from sequences of additional *Laccaria* species (Table S1) was inferred from a combined ITS and LSU dataset from GenBank, from this study and from recently generated sequences of *Laccaria* by Sheedy et al. (2013) and Popa et al. (2014). This tree provides a summary of the currently published phylogenetic information on *Laccaria* for ITS and LSU sequences. In this larger analysis (Fig. S1) containing OTUs with both ITS and LSU sequences, or just ITS or just LSU sequences, the Mexican taxon occupies a different, but also separate clade (with BS = 99) as compared to Fig. 1.

Based on the molecular information, combined with the distinctive morphological data, we propose a new species of *Laccaria* from the cloud forests of eastern Mexico.

3.2. *Taxonomy*

Laccaria roseoalbescens T. J. Baroni, Montoya & Bandala, sp. nov. Figs. 2–4.

MycoBank no.: MB809147.

Diagnosis. Differs from other *Laccaria* species by the small pileus, 7–29 mm broad, the slender stature, delicate pinkish-rose colored basidiomata with translucent striate pileus margin at first that quickly becomes opaque and white or almost white on the pileus and stipe, and the thick, distant lamellae that remain pale to bright pink and thus resembling species of *Alboleptonia*. The hymenium lacks cystidia, the pileus trama possesses thick-walled cylindrical hyphae, the basidiospores are globose, 6.5–9 μm, with large echinae, 1–2.5 μm long, and the odor of the pileus flesh when crushed is spermatic or like radish.

Holotype: MEXICO, Veracruz State, Xalapa, Santuario del Bosque de Niebla del Jardin Botanico Francisco Javier Clavijero, approximately 19°30′53.6″N, 096°56′22.9″W, 1337 m a. s. l., terrestrial under *Quercus* spp., *Carpinus caroliniana*, *Clethra, Liquidambar, Magnolia, Platanus, Turpinia*, 14 Oct 2013, leg. L. Montoya, *Montoya* 5099 (holotype, XAL).

Gene sequences ex-holotype: KJ874328 (ITS), KJ874331 (LSU).

Etymology: referring to basidiomata rosy colored (roseo) at first but becoming white (albescens).

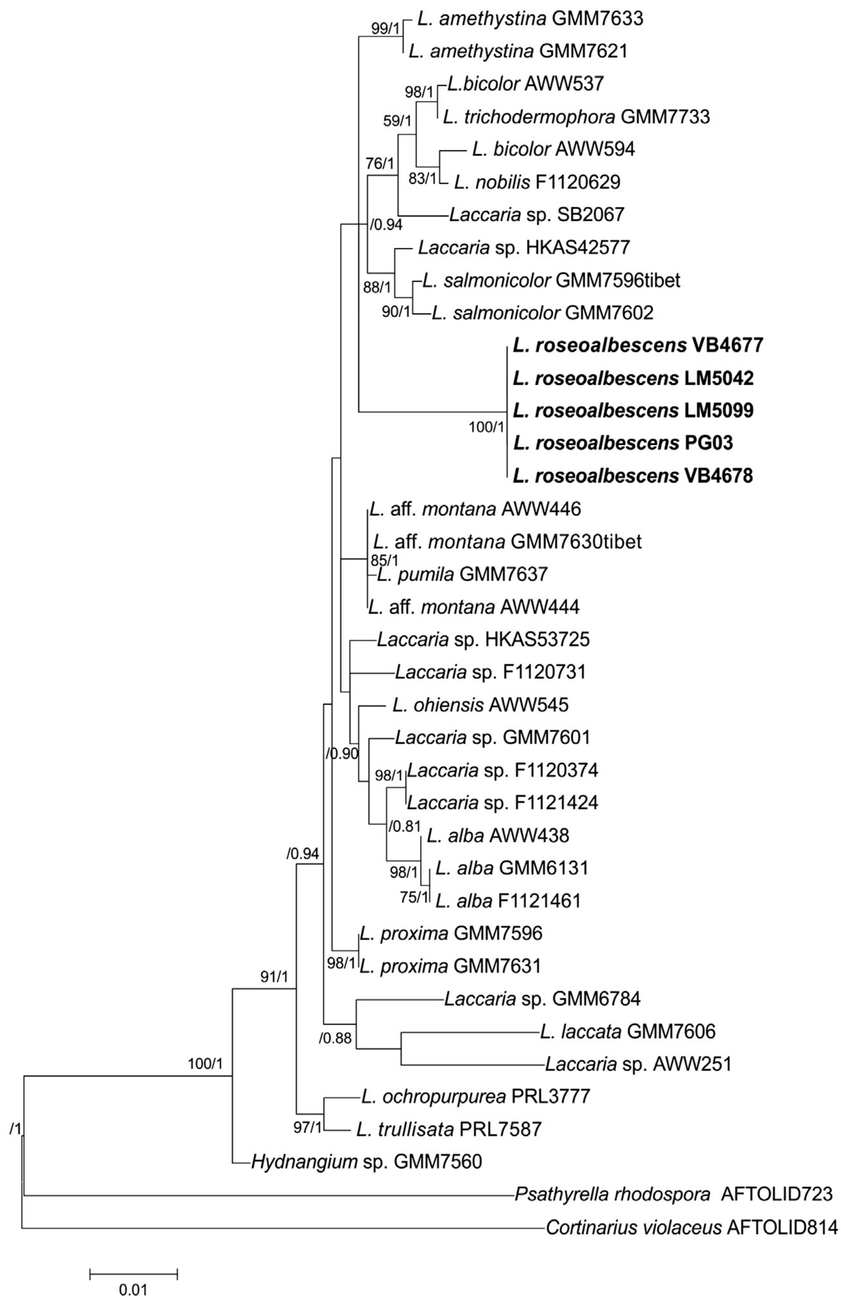

Fig. 1 – Phylogenetic relationships within *Laccaria* species inferred from the combined ITS and LSU sequence data by maximum likelihood method. Tree with the highest log likelihood (−2,978.1057), the percentage of trees in which the associated taxa clustered together (only values ≥ 70% are considered) is shown next to the branches, followed by the posterior probabilities (only values ≥ 0.90 are indicated) obtained after Bayesian inference. The tree is drawn to scale, with branch lengths measured in the number of substitutions per site.

Pileus (4–)7–29 mm broad, convex, becoming plano-convex, plane or at times uplifted when mature, round but in some cases appearing somewhat reniform from above, with or without a small broad conical umbo and frequently with shallow depressed disc with age (Fig. 2A–D), moist, opaque or faintly translucent-striate over margin at first, pale pinkish-rose or with slight buff or pinkish-flesh hues at first (7.5 YR 7/3–4, 8/2–3; 8A2, 9A3-4; 5 YR 7/3–6/4; 7.5Y8/4) in moist areas over disc or towards the margin, hygrophanous and becoming off white after losing moisture or upon drying, uneven, matted interwoven fibrillose overall, somewhat silky tomentose in the center; margin decurved when young, eventually striate or weakly sulcate or subsulcate, at times laciniate along margin, edge irregular. Context off white, ≤1 mm thick, soft. Lamellae subdecurrent or adnate, distant, with 1–3 lamellulae per lamellae, pale pinkish flesh (7B3–B4), pale pink (6–7A2) to bright fleshy-pink (10-9A2; 5 YR 7/3–6/4, 7.5Y8/4–7/4) or ochraceous (7.5YR7/3 to 2.5YR6/4), thick, broad (3 mm), edges

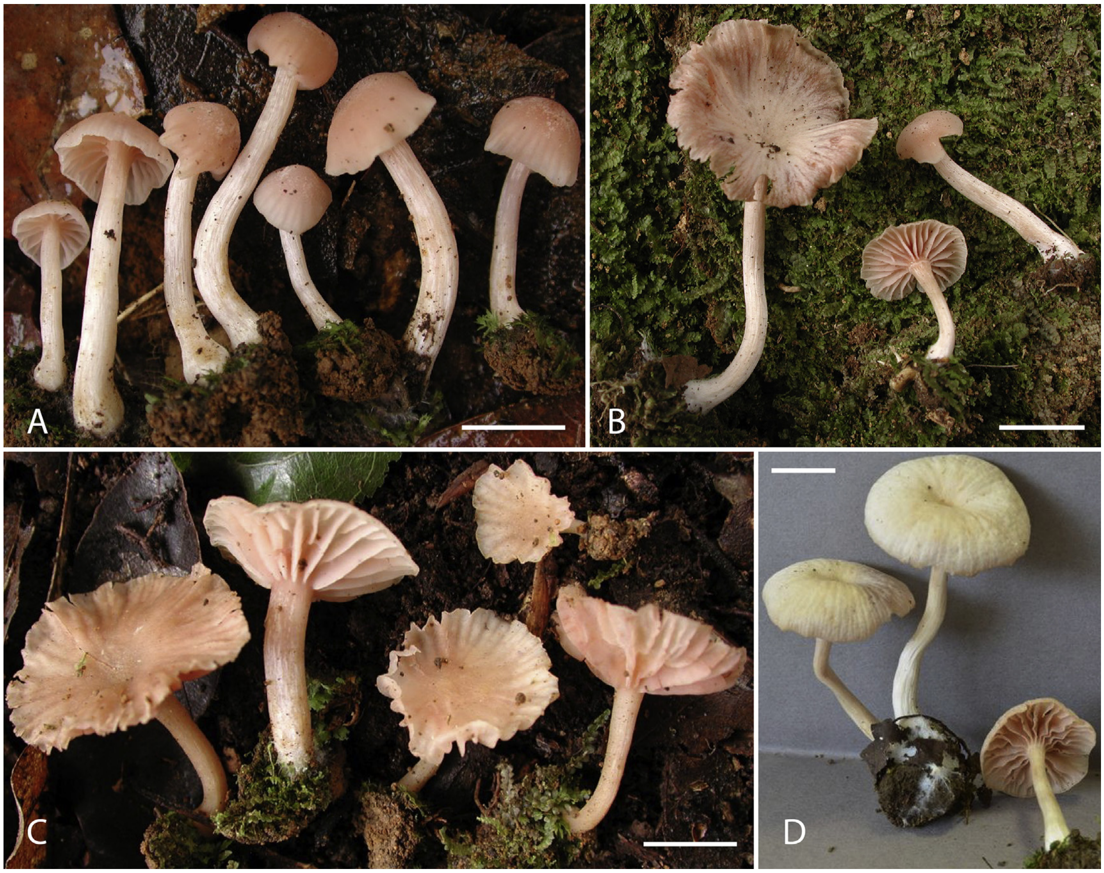

Fig. 2 – *Laccaria roseoalbescens* basidiomes. A: Gutiérrez 3. B: Montoya 5042. C: Montoya 5064. D: Baroni 10566. *Bars*: 10 mm.

smooth. Stipe 12–50 × 1–4 mm, central or eccentric, equal with an enlarged base and nearly subclavate, concolorous with pileus, pale pink (7A2), becoming whitish when loss of moisture or with age, with white mycelioid base, faintly tomentose around lamellae attachment, innate fibrous striate, fibrillose (under lens) to scurfy, splitting or fissuring, tough-pliant, narrowly hollow and white inside. Odor spermatic or weakly of radish.

Basidiospores globose (Figs. 3A, 4B,C), 6.5–9(–10) × 6.5–9(–9.5) μm, X = 7.5–8.5 × 7.5–8.3 μm, Q = 1.00–1.06 (the sporoid), with moderately large echinae 1–2.5 μm long, 0.5–1.5 μm broad at the base, thin walled, yellowish in Melzer's reagent, hyaline in 3% KOH, inamyloid, acyanophilic. Basidia 32–58 × 8–17(–19) μm, 4-sterigmate, rarely 2–3 sterigmate, clavate, with abundant often large oil droplets filling the basidia, hyaline with a yellowish contents (Figs. 3C, D, 4A). Hymenial cystidia absent, but narrowly-cylindrical, somewhat sinuous, sterile elements (17–)22–57 × 2–5(–7) μm are scattered among the clavate hymenial elements of lamellae edge (which is fertile) and occasionally dispersed along hymenium as isolated pleurocystidioid elements (Figs. 3B, C, 4A). Pileipellis a hyaline or pale straw colored layer of repent, cylindrical hyphae, 3–7(–10) μm in diam, pigments of any type completely lacking (Figs. 3E, 4A), often minutely colorless encrusted or more often inner wall of the hyphal cells with irregular thickenings projecting into the cell. Pileus trama a layer of slightly compact, radially interwoven cylindric (often short and sausage shaped) hyphae 6–13 μm diam. (65–110 μm long) and filamentose (sometimes bifurcate) hyphae 2–4 μm diam. (73–133 μm long), hyaline or straw yellow in mass, with slightly or obviously thickened walls (up to 1 μm thick) (Fig. 3F). Hymenophoral trama subregular, hyphae 3–5 μm diam., hyaline, thin walled, inamyloid. Clamp connections present on all hyphae of the basidiome.

Habit, habitat and fruiting period: scattered to subgregarious, on soil and humus on road and trail banks under mesophytic forest with *Quercus* spp., *Carpinus caroliniana*, *Clethra, Liquidambar, Magnolia, Platanus, Turpinia*, Jun through Oct.

Additional specimens examined: MEXICO, Veracruz State, Xalapa, Santuario del Bosque de Niebla del Jardin Botanico Francisco Javier Clavijero, these additional collections came from sites spread over many meters on trail banks in the area cited for the holotype, but it is not known if they are from the same or different mycelia, 30 Jul 2012, leg. T. J. Baroni, *Baroni 10566* (CORT); 25 Jun 2013, leg. M. Herrera, *Herrera* 55; 25 Jul 2013 *P. Gutiérrez* 3; 2 Aug 2013, leg. V. Bandala, *Bandala 4677, 4678*; 5 Aug 2013, leg. L. Montoya, *Montoya 5042*; 19 Aug 2013, leg. L. Montoya, *Montoya 5064*; 24 Sep 2013, leg. J. C. Corona, *Corona 899*; 3 Apr 2014, leg. V. Bandala, *Bandala 4692, 4693, 4694*, 4695 (all except *Baroni 10566* at XAL).

4. Discussion

Laccaria roseoalbescens is distinctive by the following set of characteristics: the small stature of the basidiomes, the fragile

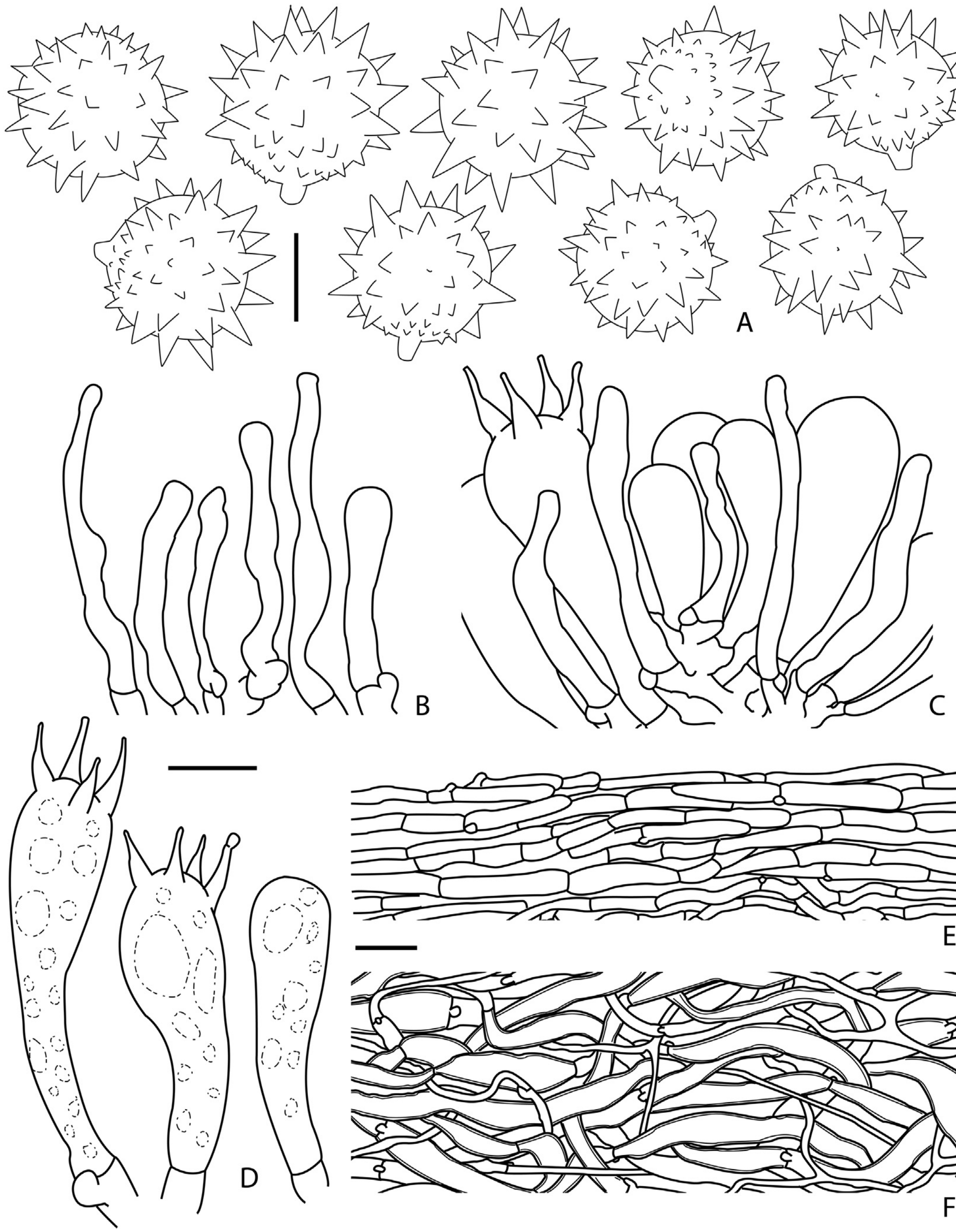

Fig. 3 – *Laccaria roseoalbescens* (Montoya 5099, holotype) line drawings. A: Basidiospores. B: Sterile elements in the hymenium. C: Basidium, basidioles and sterile elements at tip of lamellae (tangential section). D: Basidia and basidiole. E: Pileipellis. F: Pileus trama. *Bars*: A 5 μm; B–D 10 μm; E, F 25 μm.

and slender habit with thin pileus that is usually translucent towards the margin when fresh, the colors ranging from pinkish-flesh hues when fresh but becoming almost white and opaque with loss of moisture (the basidiome lacking any brownish, reddish, violaceous or purplish tinges), the context with spermatic or weakly radish smell; the pileus trama with filamentous and cylindrical, thick-walled hypha; the spherical spores with large echinae, the lamella edges fertile and lacking cheilocystidia and the large, 4-spored basidia. When the first collection was found in 2012, the hygrophanous basidiomes resembled species of *Alboleptonia* in the field because of their delicate stature, white pileus and stipe with the contrasting dark pinkish, broad and distant lamellae. However, the globose strongly echinate basidiospores under the light microscope quickly rectified that erroneous field identification.

Laccaria roseoalbescens superficially resembles the Asian *L. alba* Zhu L. Yang & Lan Wang when the basidiomata of *L.*

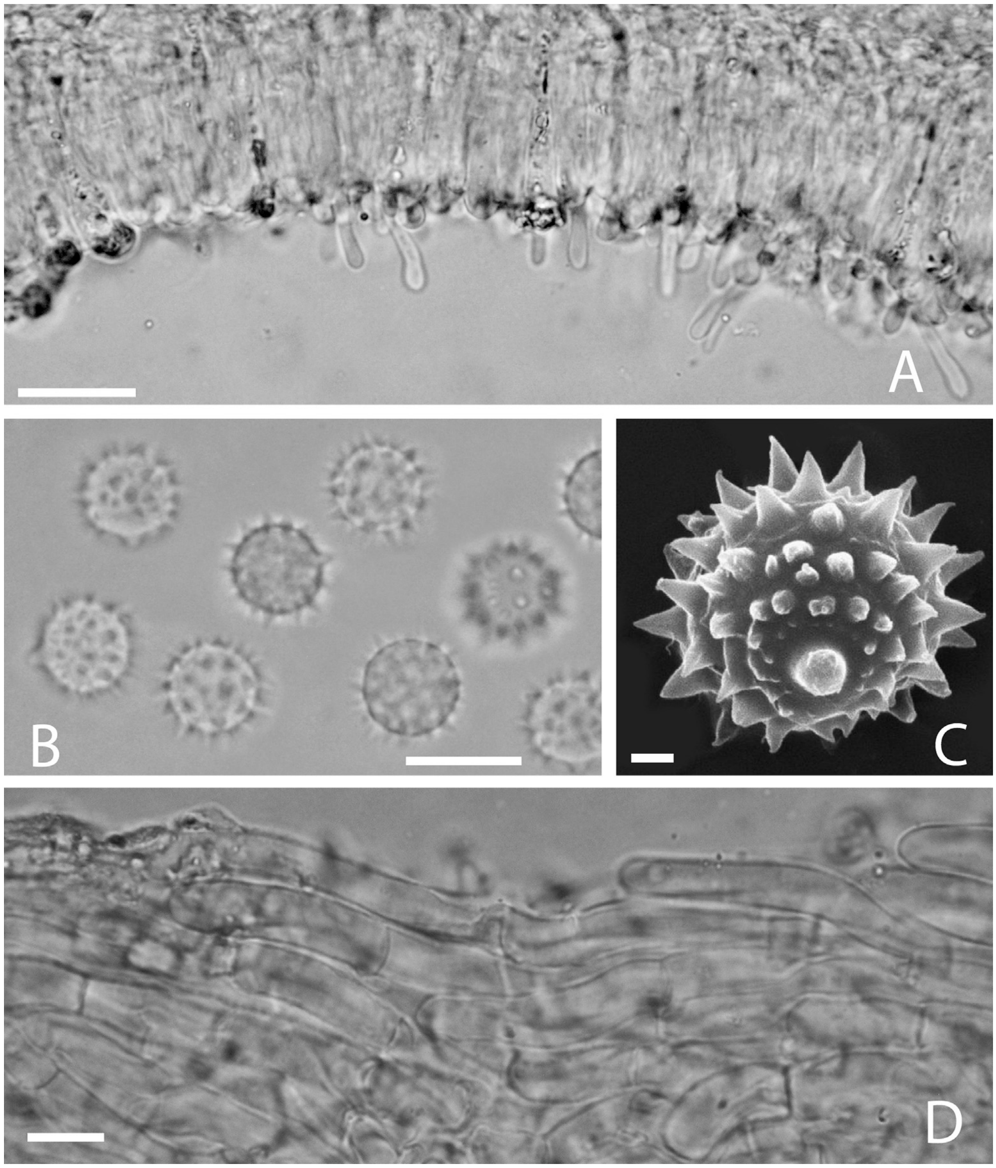

Fig. 4 – *Laccaria roseoalbescens*. A: Elements of hymenium in tangential section (Montoya 5099, holotype). B–C: Basidiospores (Bandala 4678). D: Pileipellis (Montoya 5064). *Bars*: A 25 μm; B, D 10 μm; C 1 μm.

roseoalbescens have lost moisture and become nearly white. Phylogenetically these two taxa are distantly related (Fig. 1). *Laccaria alba* is a species described from China, growing under *Quercus* and *Lithocarpus*, and differs from the Mexican taxon by its more robust appearance and rigid consistency, indistinct smell, subsulcate pileus margin, pileus trama with tightly interwoven, filamentous, thin walled hyphal elements 3–16 μm diam. and cylindrical or subcapitate cheilocystidia (Wang et al. 2004). The inconsistent occurrence of sterile elements along the lamellae edges in the Mexican materials suggest these are not true cheilocystidia, thus we consider *L. roseoalbescens* without cheilocystidia. The sterile cells on the lamella edges in *L. roseoalbescens* are cylindrical in contrast to the clavate basidia and basidioles and project only slightly beyond the hymenium. These sterile cells can also appear irregularly distributed on the lamella faces (e.g. Figs. 3B, C, 4A).

Laccaria roseoalbescens appears to be phylogenetically isolated from other taxa in the genus (Fig. 1). It is distinct from *L. aurantia*, *L. fulvogrisea* and *L. yunnanensis*, recently described by Popa et al. (2014) from China, and isolated from the potential new Australian species in the phylogenies obtained by Sheedy et al. 2013 (see Supplementary Fig. S1). The lack of violaceous or purple pigments in the basidiomata or the mycelium at the base of the stipe would place *L. roseoalbescens* in Metasection *Laccaria* (Mueller 1992) using morphological data. However, *L. roseoalbescens* comes out on the same branch system with species assigned to Metasection *Amethystina*, i.e. those taxa with violet mycelium at the base of the stipe (Mueller 1992). Wilson et al. (2013) described *L. salmonicolor* from Xizhang, China, also a species lacking violet basal mycelium, but clustering phylogenetically with taxa that do produce such colorful mycelium over the base of the stipe (Fig. 1). Even though both of these taxa cluster phylogenetically with species that uniformly produce violet basal mycelia, they are not similar morphologically and do not appear to be closely related (Fig. 1). At this time, the phylogenetic relationship of *L. roseoalbescens* to other taxa in *Laccaria* is not obvious.

Morphologically *L. roseoalbescens* is similar to *L. striatula* (Peck) Peck, due to its stature, translucent-striate pileus margin (a feature not common for *Laccaria* taxa), the globose basidiospores with rather large echinae and the lack of cheilocystidia. However, *L. striatula* differs markedly from *L. roseoalbescens* by its reddish brown to orange-brown pileus colors that do not become white with age, the larger basidiospores [7–10(–12) × 7–10(–12) μm; X = 8–9(–10) × 7.7–9(–10) μm], the wider hyphae in the lamella trama [3–11(–27) μm] and the pileipellis with hyphal fascicles with terminal cells 33–85 × 6–18.5 μm (Mueller 1992). One of us (T. J. Baroni) has collected *L. striatula* many times in northeastern North America and knows this species very well. Even though they share some similarities, they are very different species morphologically.

Disclosure

The authors declare no conflicts of interest. All the experiments undertaken in this study comply with the current laws of the country where they were performed.

Acknowledgments

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Appendix A. Supplementary data

Supplementary data related to this article can be found at http://dx.doi.org/10.1016/j.myc.2015.06.002.

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